4.9 Holditch's Theorem

WORKSHEET

The question

STEP III 2011, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q5

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Check the theorem in the simplest case. If $\mathcal{C}$ is a circle of radius R, what curve does P trace, and what is the area of the ring?

STAGE 1 Area from coordinates

Covers: the area formula

Ask yourself With $x = r\cos\theta$ and $y = r\sin\theta$, all functions of t, find $\dot{x}$ and $\dot{y}$, and compute $x\dot{y} - y\dot{x}$.

STAGE 2 The ends of the rod

Covers: the coordinates of A and B

Ask yourself The rod points in the direction $(\cos t, \sin t)$ from A to B. How far is P from each end?

STAGE 3 The area swept by OA

Covers: the area swept by OA

Ask yourself Apply $(*)$ to A. Expand $x_A\dot{y}_A - y_A\dot{x}_A$ and sort the terms by powers of a.

Pause and try Integrate each group of terms from 0 to 2π. For $[B]$: B is given by the formula for A with a replaced by what?

STAGE 4 Eliminating f

Covers: the area between $\mathcal{C}$ and $\mathcal{D}$

Pause and try Both A and B go once round the same curve $\mathcal{C}$. What does that tell you about $[A]$ and $[B]$? Use it to find f.

What's really going on

Green's theorem and the planimeter

The formula $(*)$ is a special case of *Green's theorem*, which relates an integral round a closed curve to an integral over the region inside it. In its area form, it says that the area enclosed by a curve is $\frac{1}{2}\oint(x\,\mathrm{d}y - y\,\mathrm{d}x)$, an integral that can be computed by travelling round the boundary alone. Engineers exploited this with the *planimeter*, a mechanical instrument invented in the nineteenth century (Jakob Amsler's 1854 design was the most successful): you trace the boundary of a shape on a map or diagram with a pointer, and a wheel records the enclosed area. Planimeters were used to measure land areas, to compute the work done in steam engines from indicator diagrams, and in many other applications, until digital methods replaced them. Holditch's theorem is closely related: the rod is like the arm of a planimeter.

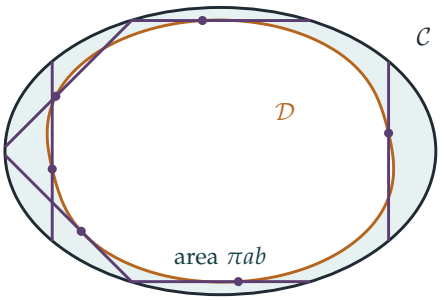


Figure 4.9.1 Holditch's theorem for an ellipse with semi-axes 3 and 2, and a rod of length 2.5 with $a = 1$, $b = 1.5$. Some positions of the rod are shown (plum), with the point P marked. The curve $\mathcal{D}$ traced by P (amber) was computed numerically, and the shaded ring between the curves has area $4.7124\ldots = 1.5\pi$, exactly as the theorem predicts.

The fine print

Holditch stated his theorem without much care for conditions, and it needs some. The rod must be able to go all the way round with both ends on the curve, and the traced curve $\mathcal{D}$ must be a simple closed curve traced once; if the rod is too long compared with the width of $\mathcal{C}$, $\mathcal{D}$ can cross itself, and the "area" given by the formula counts some regions twice or with a negative sign. The question avoids these difficulties by assuming $\mathcal{D}$ is a convex curve containing O. Finding exactly when the theorem holds is a good example of how a beautiful idea must be made precise, a theme of Chapter 3.

Try this next

1 **The area of an ellipse.** Use $(*)$ with $x = p\cos t$, $y = q\sin t$ to show that an ellipse with semi-axes p and q has area πpq.

2 **The circle case.** Prove the intersecting chords theorem: if two chords of a circle meet at P, the products of the lengths of their two segments are equal. Deduce Holditch's theorem for a circle.

3 **The midpoint.** Show that $f = \pi(a - b)$ means that when P is the midpoint of the rod, $f = 0$. What is the ring's area then, in terms of the rod's length L?

4 **A triangle.** HARDERUse $(*)$ to find the area of the triangle with vertices $(0,0)$, (x_1, y_1), (x_2, y_2), traversing its boundary, and show it equals $\frac{1}{2}|x_1y_2 - x_2y_1|$. This is the start of the "shoelace formula" for the area of any polygon.

Further reading

- **Mark Levi, *The Mathematical Mechanic*** (Princeton University Press). Uses physical reasoning, with rods, wheels and springs, to prove mathematical results, in the spirit of this section.
- **H. M. Schey, *Div, Grad, Curl, and All That*** (Norton). An informal introduction to vector calculus, including Green's theorem.
- **Hamnet Holditch, "Geometrical theorem"**, *Quarterly Journal of Pure and Applied Mathematics* 2 (1858). The original note, a single short paragraph with none of the fine print.

The full guided solution is in Section 4.9:
https://beyondstep.keithmcnulty.org/ch04/holditch.html